



MILESTONE 1 · DUE 10 AUGUST 2026 · 20%

Project Charter

SAS821S CAPSTONE PROJECT · EXACTLY TWO GROUP MEMBERS

1. Group and project identification

Field	Group response
Topic ID and approved topic	T18 · Insurance Claims Fraud, Identity Abuse and Data-Leakage Analytics
Working project title	Insurance Claims Fraud Detection and Identity Risk Analytics
Member A	223127744 · Shanon Chiwara · Data and Modelling Lead]
Member B	216069076 · Ephath Shimhanda · Security Engineering and Intelligence Lead]
Repository/location	Github
Date submitted	10/08/26



2. Problem and decision statement

Insurance companies process large volumes of claims and sensitive customer information every day, making them potential targets for fraud, identity abuse and insider misuse. In the identified scenario, suspicious insurance claims have been observed following changes to customers' identity or KYC profiles, unusual document uploads and overrides performed by staff members. These activities may represent isolated incidents, but they may also indicate coordinated fraudulent behaviour involving claimants, compromised accounts, devices, documents or employees. If these relationships are not identified early, fraudulent claims may be approved, resulting in financial losses, while sensitive claimant information could also be accessed or misused without authorization.

The problem affects several stakeholders, including the insurer, claims and fraud investigation teams, employees with access to customer information, legitimate policyholders and management. Claims and fraud investigators need to determine whether suspicious activities are genuine fraud or legitimate customer behaviour. Security teams need to identify unusual authentication, device and staff-access patterns that could indicate account compromise or insider risk. Management also needs reliable evidence to make decisions about whether a claim should be approved, rejected or escalated for further investigation.

The main challenge is that suspicious activity may not be visible when each data source is examined separately. A change in a claimant's identity profile, for example, may appear normal on its own, while the same change followed by a new device login, unusual document upload and staff override may indicate a higher-risk event. Therefore, analysing relationships across claims, identity changes, authentication records, devices, documents and staff access is necessary to identify connected patterns of suspicious behaviour.

The decision that the proposed analytics solution must support is whether a claim or related activity should be considered low, medium or high risk and what action should follow. High-risk cases should be prioritised for investigation, while legitimate low-risk claims should continue through normal processing. The solution should also help identify potential insider misuse, suspicious access to claimant information and weaknesses in fraud detection models. Ultimately, the objective is to provide investigators and management with evidence-based risk indicators that support faster and more accurate decisions while reducing financial losses and protecting sensitive claimant information.

3. Scope and exclusions

In scope	Out of scope
Claims records, identity/KYC changes, portal authentication logs, device information, document metadata, OCR text, staff access records and investigation notes.	Live penetration testing of insurance systems.
Analysis of these datasets to identify suspicious claims and possible fraud.	Real-world verification of customer identities.
Fraud classification using machine learning techniques.	Manual inspection of physical documents.
Access analytics to detect unusual login and account activity.	Legal action, disciplinary action or prosecution of suspects.
Entity linkage analysis to identify relationships between claims, users, devices and employees.	Recovery of financial losses caused by fraud.
Text mining of uploaded documents and investigation notes.	Monitoring activities outside the insurer's available datasets.
Insider-risk detection and behavioural anomaly analysis.	Fully automated approval or rejection of claims.
Development of a decision-support prototype that provides risk scores, alerts and recommendations to investigators.	Full deployment of the solution in a production environment.



4. Objectives and security analytics questions

Objective/question	Measure of completion
Objective 1: Identify and classify insurance claims that show signs of possible fraud using the available claims, identity, login, device and document data.	The analysis produces a list of suspicious claims and shows how well the classification method identifies them using measures such as precision, recall and F1-score.
Objective 2: Find links between suspicious claims, identity changes, devices, uploaded documents and staff activities that may indicate coordinated fraud or insider misuse.	The analysis identifies suspicious connections or unusual patterns between the different data sources and provides evidence for the identified cases.
Objective 3: Test how well the fraud detection method performs when it is given unusual or manipulated data.	The model is tested using simulated suspicious or changed inputs, and the effect on its results is recorded.
Question 1: What patterns are commonly found in claims that appear to be suspicious or fraudulent?	The analysis identifies the main factors and patterns associated with suspicious claims using statistics, visualisations and model results.
Question 2: Are suspicious claims connected to identity changes, unusual devices, document uploads or staff access?	The different datasets are compared and linked to identify unusual relationships and activities around suspicious claims.
Question 3: How well can the proposed approach detect suspicious claims, including cases where the data has been manipulated?	The model's results are evaluated using measures such as false positives, false negatives, precision, recall and F1-score before and after testing.

5. Learning Sessions 1-10 coverage matrix

Session	Course area	Planned project evidence
1	Foundations of security analytics	Define the insurance fraud problem, key stakeholders, security risks and decision requirements. Evidence will include the problem statement, project scope, objectives and analytics questions.
2	Analytical software and tools	Use Python and related libraries to clean, transform, integrate and visualise insurance datasets. Evidence will include scripts, notebooks, data-cleaning logs and visualisations.
3	Machine-learning fundamentals	Apply supervised learning techniques to distinguish potentially fraudulent claims from legitimate claims. Evidence will include feature selection, model training, testing and performance evaluation results.
4	Machine learning in cybersecurity	Build and evaluate a fraud-risk classification model using claim, identity, authentication, device and document-related features. Evidence will include model outputs, confusion matrices and performance metrics.
5	Analytics and incident response	Analyse suspicious claims and related activities to support investigations and response decisions. Evidence will include risk assessments, investigation findings and incident timelines.
6	Security intelligence	Correlate information from multiple data sources to identify emerging fraud patterns and coordinated activity. Evidence will include intelligence summaries, entity-linkage findings and risk indicators.
7	Access analytics	Analyse authentication records, device activity and staff access logs to identify unusual behaviour and potential insider misuse. Evidence will include behavioural baselines, anomaly reports and access-analysis results.
8	Simulations and security processes	Create simulated fraud and misuse scenarios to test how the proposed solution responds to suspicious activity. Evidence will include scenario comparisons, simulation outputs and documented findings.
9	Security and text mining	Analyse document metadata, OCR text and investigation notes to identify suspicious documents, repeated information and unusual text patterns. Evidence will include extracted indicators, text-analysis outputs and visualisations.



Session	Course area	Planned project evidence
10	Advanced security intelligence	Combine findings from claims, identity, device, document and access data to produce risk assessments and recommendations. Evidence will include predictive risk scores, adversarial testing results and final recommendations.

6. Proposed data sources

Source	Format/volume	Key fields	Provenance/access	Security/ethics controls
Claims and Identity/KYC Records	CSV/Excel; estimated volume based on available dataset	Claim ID, Claimant ID, Claim Date, Claim Type, Claim Amount, Claim Status, Identity Changes, KYC Update Date	Synthetic or approved academic dataset	Personal identifiers will be removed or anonymised before analysis. Data will be used only for academic purposes and stored securely.
Portal Authentication and Device Data	CSV and log files; event-based records	User ID, Login Time, IP Address, Device ID, Authentication Result, Location/Network Information	Synthetic or approved security-log dataset	User identifiers will be anonymised where necessary. Access to the data will be limited to project members and used only for analysis purposes.
Staff Access and Investigation Records	CSV and log files; event-based records	Staff ID, Access Time, Record Accessed, Action Performed, Override Activity, Investigation Outcome	Synthetic or approved organisational dataset	Staff identities will be anonymised and findings will be reported objectively. Results will not be used to make unsupported accusations against individuals.
Document Metadata and OCR/Text Data	PDF metadata, CSV files and text records; 200+ text records	Document ID, Claimant ID, Upload Time, File Type, File Size, OCR Text, Document Similarity Indicators	Synthetic or approved document dataset	Sensitive information contained in documents will be removed or masked where necessary. Text data will be analysed only for approved academic and research purposes.

7. Proposed analytical methods

Component	Proposed method/tool	Expected output
Baseline/EDA	Data cleaning, descriptive statistics, missing-value analysis, distribution analysis and correlation analysis using Python.	Cleaned dataset, summary statistics, visualisations and identification of unusual patterns, trends and outliers.
Supervised ML	Classification using Logistic Regression and Random Forest.	Fraud-risk predictions for insurance claims, evaluated using accuracy, precision, recall, F1-score and ROC-AUC.
Unsupervised or UBA	Clustering and anomaly detection using claimant, device and access-behaviour data.	Identification of unusual users, devices, claims or access patterns that differ from established behaviour baselines.
Incident investigation	Timeline analysis and correlation of claims, identity changes, authentication events, devices, documents and staff activities.	Investigation timelines, correlated events and supporting evidence linking suspicious activities.
Security intelligence	Entity-linkage analysis and risk indicators across multiple datasets.	Identification of connected entities, suspicious relationships and prioritised high-risk cases for investigation.



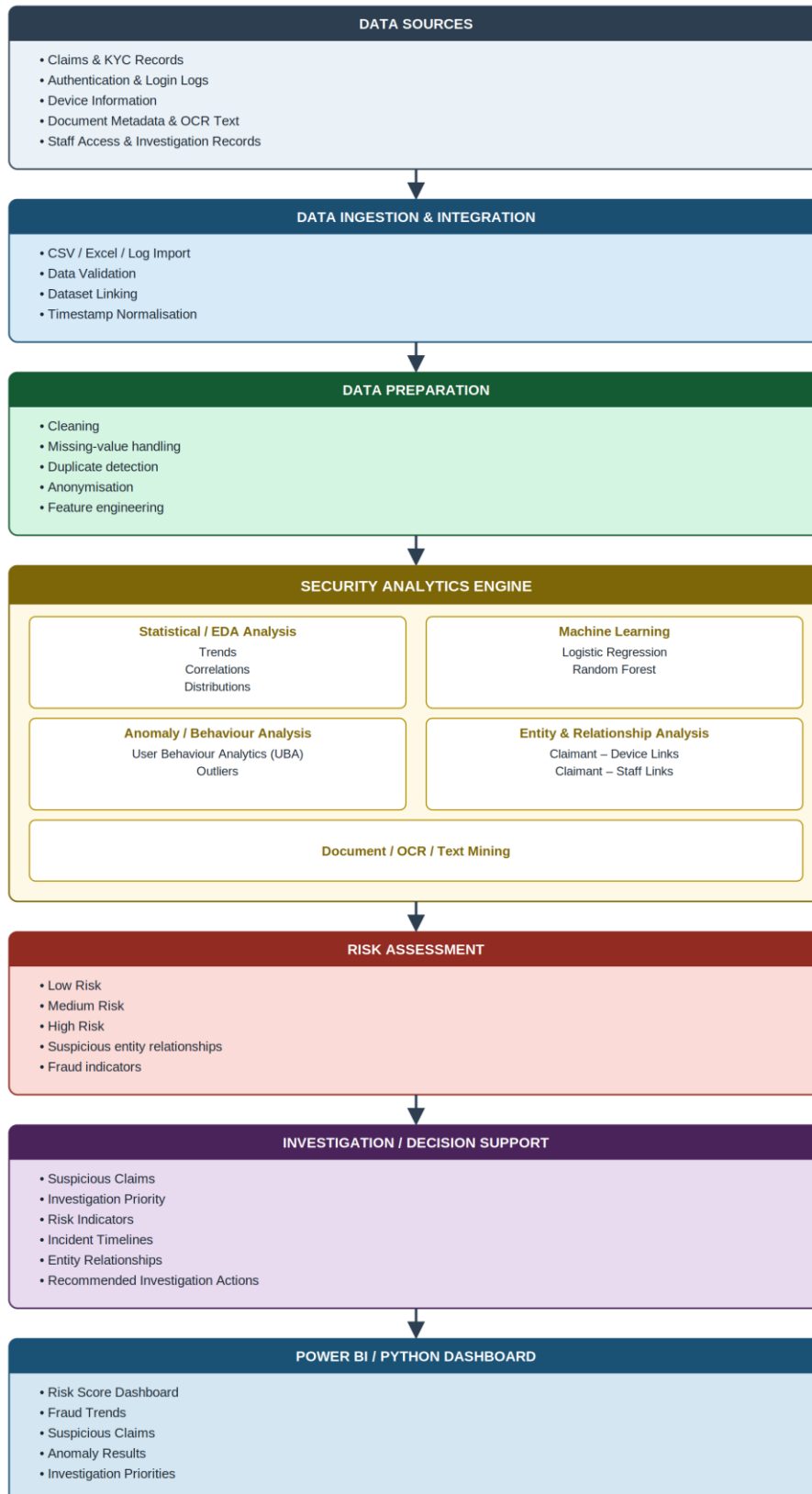
Component	Proposed method/tool	Expected output
Simulation	Simulated fraudulent claims, unusual access scenarios and manipulated inputs.	Comparative results showing how the analytics approach responds to different suspicious activity scenarios.
Text mining/NLP	OCR text cleaning, keyword analysis, frequency analysis and document-similarity analysis.	Suspicious keywords, repeated content, similar documents and other unusual text patterns.
Predictive/adversarial analysis	Model sensitivity testing using altered or manipulated claim, identity and access-related features.	Comparison of model performance before and after manipulation, including changes in risk scores and detection accuracy.
Prototype/dashboard	Power BI, Python-based dashboard or similar decision-support interface.	Interactive display of risk scores, suspicious claims, entity relationships, trends and investigation priorities.

8. High-level architecture

The proposed system will follow a structured security analytics pipeline that integrates insurance claims, identity/KYC, authentication, device, document and staff-access data. The architecture is designed to move from raw evidence collection through data preparation and analysis to risk scoring and investigation support.



Figure 1: Proposed Security Analytics Architecture





9. Roles and RACI

Workstream	Member A	Member B	Review/approval evidence
Data acquisition and quality	A/R	C	Dataset inventory, cleaning logs, data quality checks and peer review records
ML and anomaly modelling	A/R	C	Model outputs, evaluation metrics, experiment records and code review
Investigation and access analytics	C	A/R	Access-analysis findings, investigation timelines and validation notes
Simulation and intelligence	C	A/R	Simulation results, intelligence summaries and review comments
Prototype and testing	R	R	Test cases, dashboard reviews, defect log and testing results
Documentation and presentation	R	R	Shared report drafts, presentation slides, meeting notes and final approval by both members

10. Work plan

Task	Owner	Start	Finish	Dependency	Output
Task	Owner	Start	Finish	Dependency	Output
Task 1: Finalise project scope, objectives and research questions	Both	10/08/2026	11/08/2026	Approved project topic	Final project scope and objectives
Task 2: Identify, obtain and document datasets	A/R: Shanon C: Ephath	10/08/2026	14/08/2026	Task 1	Dataset inventory and provenance records
Task 3: Data cleaning, anonymisation and quality assessment	A/R: Shanon C: Ephath	12/08/2026	18/08/2026	Task 2	Clean datasets, data-quality report and cleaning logs
Task 4: Exploratory data analysis and visualisation	A/R: Shanon C: Ephath	16/08/2026	22/08/2026	Task 3	Statistical analysis, graphs and initial fraud indicators
Task 5: Machine-learning and anomaly-detection development	A/R: Shanon C: Ephath	20/08/2026	28/08/2026	Task 5	Classification models, anomaly results and evaluation metrics
Task 6: Investigation correlation, security intelligence, dashboard and reporting	A/R: Ephath C: Shanon	25/08/2026	05/09/2026	Task 5- 6	Investigation timelines, risk indicators, dashboard, recommendations and final report



11. Risk and ethics register

Risk	Likelihood	Impact	Mitigation	Owner
Data unavailable or unsuitable	Medium	High	Use approved synthetic or academic datasets where necessary; document data limitations and avoid making unsupported conclusions.	A/B
Privacy or sensitive-data exposure	Medium	High	Remove or anonymise personally identifiable information, restrict access to project members and store datasets securely.	A/B
Model bias or misleading results	Medium	High	Evaluate precision, recall, F1-score, false positives and false negatives. Compare model outputs and document limitations. Human review will remain necessary for high-risk cases.	A/B
Scope or time overrun	Medium	Medium	Follow the agreed work plan, prioritise core objectives and conduct regular progress reviews between members.	A/B
Technical integration failure	Medium	Medium	Maintain backups, use common data formats such as CSV, document dependencies and test each analytical component separately before integration.	A/B
Unequal contribution between group members	Low	Medium	Follow the RACI allocation, retain evidence of individual contributions and conduct peer reviews of major deliverables.	A/B
Incorrect or incomplete analytical results	Medium	High	Validate datasets, cross-check results using different analytical techniques and document assumptions and limitations.	A/B
False accusation of fraud or insider misuse	Low	High	Treat analytical results as risk indicators rather than proof of wrongdoing. Suspicious cases must be reviewed by an authorised human investigator.	A/B
Unauthorised use of real customer information	Low	High	Use synthetic or formally approved datasets. No real customer information will be included in screenshots, demonstrations or public repositories.	A/B
Academic integrity or inappropriate AI use	Low	High	Properly acknowledge external sources, code and permitted AI assistance. Verify all AI-generated content and ensure that the final work represents the group's own understanding.	A/B

12. Responsible AI and academic integrity statement

AI tools may be used for brainstorming, explaining concepts and improving writing. External code, datasets, and references will be properly acknowledged and cited. The group will verify all AI-generated and externally sourced content before use and disclose permitted AI assistance. All final work and findings remain the responsibility of the group.

13. Group agreement and declaration

- We confirm that this group consists of exactly two students.
- We accept the role allocation and will retain evidence of our contributions.
- We understand that both members must explain the complete solution during the presentation and Q&A.
- We will use only authorised data and will not test live third-party systems.
- We will disclose permitted AI assistance, reused code and external data sources.

Member	Signature	Date
Shanon Chiwara	S.C	09/08/2026
Ephath Shimhanda	E.H	10/08/2026

14. Lecturer approval

Decision	Comments/conditions	Lecturer/date